SEMESTER 5 | 4-WEEK STATUS UPDATE | OCT 8 - NOV 5, 2026

# AccessGuard Health

Rubrics progressed, directory and cloud identity caught up, and Azure Key Vault planned

Sabitha S Karanavar | DAE 12-month Cohort | Semester 5 | Healthcare IAM portfolio project

- 11 of 12 rubric items done
- AD DC + Entra ID caught up
- Azure Key Vault planned
- Schoology videos watched
- FieldDAE attended

GitHub: github.com/sabi-s/dae-6-month-Sabi

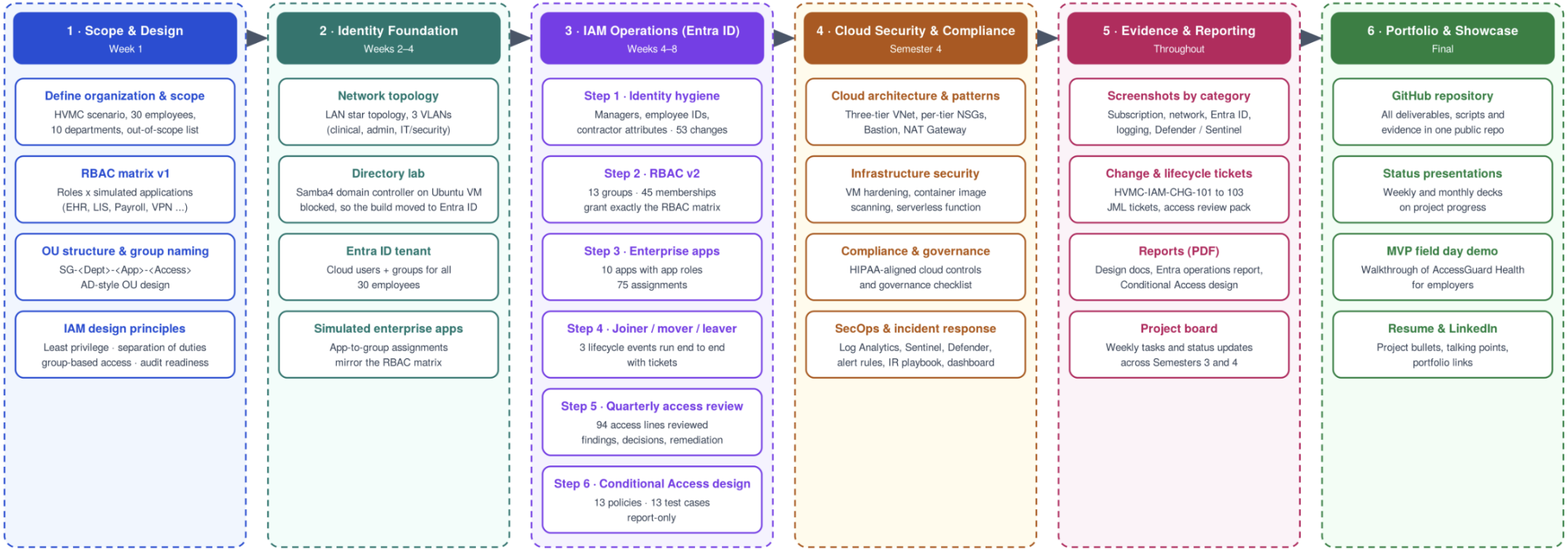
PROJECT OVERVIEW

# Keeping hospital records safe with identity and access management

Every hospital has to answer one question: who should see patient information, and who should not? I built a system that answers it for every person, from their first day to their last.

## AccessGuard Health — Project Workflow

Horizon Valley Medical Center (HVMC) · 30 employees · 10 departments · 12-week build, from design to a published portfolio



# Too much access puts patient data at risk

### Sensitive data

Patient records hold medical history and personal details that can't be changed once exposed.

### Strict rules

HIPAA requires hospitals to limit who sees patient information, and to prove it with records and audits.

### Constant change

Shift workers, rotating staff and contractors join, move and leave all the time.

The weak spot is often an account with more access than the job needs, or one that was never shut off. That is the gap AccessGuard Health closes.

WHAT I BUILT

# Right access, right person, right time



01

## Access by role

Least privilege in Entra ID: each role gets only the access it needs.



02

## Account lifecycle

Created on day one, updated on a role change, shut off when people leave. Contractors expire automatically.



03

## Layers of protection

Three isolated network zones, Azure Bastion for admins, and multi-factor sign-in.



04

## Rules and readiness

Controls mapped to HIPAA and NIST, plus a tested incident-response playbook.

PROGRESS SUMMARY

# What I completed in the last 4 weeks

## Risk Management rubric

Completed, documented and pushed to GitHub.

DONE

## Cryptography rubric

5 of 6 criteria completed and pushed to GitHub; Configure Cryptographic Protocols in progress.

5 OF 6 DONE

## Active Directory DC work

Caught up on all domain controller work; completed and pushed to GitHub.

DONE

## Entra ID work

Caught up on pending Entra ID work and pushed to GitHub. Now in the final stage of Conditional Access, documenting everything.

DONE

## New tool: Azure Key Vault

Chosen for the project; plan written and uploaded to the learning folder.

PLANNED

## FieldDAE event

Attended the FieldDAE event at DAE.

DONE

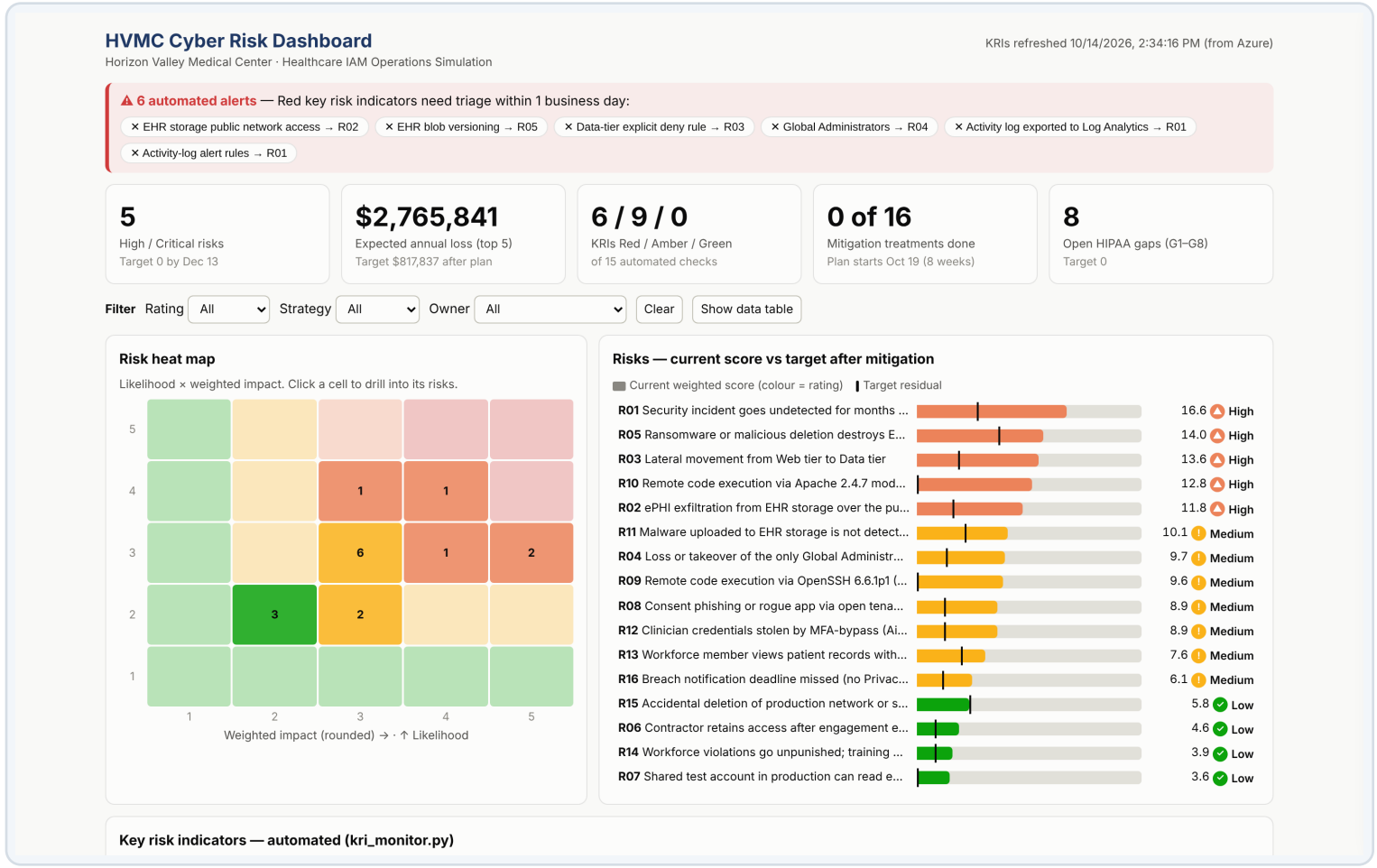
RISK MANAGEMENT

# Risk Management rubric: all items done

| # | Rubric item | Status | One-line summary | GitHub |
|---|---|---|---|---|
| 1 | **Risk Management Fundamentals** | DONE | NIST RMF/ISO 31000 risk framework, governance and escalation, and a HIPAA compliance matrix. | **Pushed** |
| 2 | **Risk Assessment Techniques** | DONE | Three identification methods, 5x5 matrix, EMV and ALE analysis, and a scored risk register. | **Pushed** |
| 3 | **Risk Evaluation Methods** | DONE | Weighted multi-criteria scoring model plus Business Impact Analysis (RTO, RPO, MTD). | **Pushed** |
| 4 | **Risk Mitigation Planning** | DONE | NIST CSF/ISO 27001 controls, all four treatment options, and residual risk calculated. | **Pushed** |
| 5 | **Risk Communication** | DONE | Executive risk report, stakeholder communication plan, risk dashboard and KPI framework. | **Pushed** |
| 6 | **Risk Management Implementation** | DONE | Program plan with phased rollout, framework gap analysis and continuous monitoring. | **Pushed** |

*All 6 of 6 rubric criteria completed; documentation pushed to GitHub.*

# HVMC Cyber Risk Dashboard




*Screenshot: HVMC_Risk_Dashboard.html from my risk communication deliverable*

## What it shows

- Risk heat map: likelihood against weighted impact for each risk
- 16 risks scored, each with a target after mitigation
- 15 automated key risk indicators that raise alerts
- HIPAA gap tracking alongside the risk register

CRYPTOGRAPHY

# Cryptography rubric: 5 of 6 done, 1 in progress

| # | Rubric item | Status | One-line summary | GitHub |
|---|---|---|---|---|
| 1 | **Demonstrate Cryptographic Fundamentals** | DONE | Python report on 2 cryptographic primitives with math principles and security analysis. | **Pushed** |
| 2 | **Implement Symmetric Cryptography** | DONE | AES in ECB and CBC modes, key management, a stream cipher and best practices. | **Pushed** |
| 3 | **Apply Hash Functions and Integrity** | DONE | Modern password hashing, HMAC authentication and a file integrity checker. | **Pushed** |
| 4 | **Utilize Public Key Cryptography** | DONE | RSA key pairs, asymmetric encryption, a simple PKI and secure key exchange. | **Pushed** |
| 5 | **Implement Digital Signatures** | DONE | RSA signatures, certificate management, verification process and trust model. | **Pushed** |
| 6 | **Configure Cryptographic Protocols** | WORKING ON IT | TLS on a web server, SSH key-based login, secure channel and Wireshark analysis. | **In progress** |

*5 of 6 rubric criteria completed and pushed to GitHub; Configure Cryptographic Protocols is in progress.*

# Public key cryptography: 19 of 19 tests pass

```
cryptography_1/public_key_crypto: python3 test_and_demo.py

HVMC Public Key Cryptography - test and demo run
Python 3.9.6 on Darwin arm64, 2026-10-26 13:37; cryptography package: 47.0.0; LibreSSL 3.3.6

=== Unit tests ===
test_crt_and_blinding_match_plain_decryption (__main__.TestRSA) ... ok
test_keygen_2048_properties_and_library_accepts (__main__.TestRSA) ... ok
test_number_theory (__main__.TestRSA) ... ok
test_oaep_interop_with_library_both_ways (__main__.TestRSA) ... ok
test_oaep_round_trip_random_and_limits (__main__.TestRSA) ... ok
test_pkcs1v15_verify_on_real_certificates (__main__.TestRSA) ... ok
test_textbook_example (__main__.TestRSA) ... ok
test_encrypted_storage_permissions_and_passphrase (__main__.TestKeyManagement) ... ok
test_expiry_enforced (__main__.TestKeyManagement) ... ok
test_hybrid_rotation_and_lifecycle (__main__.TestKeyManagement) ... ok
test_chain_valid_mine_and_library (__main__.TestPKI) ... ok
test_csr_policy (__main__.TestPKI) ... ok
test_rejections (__main__.TestPKI) ... ok
test_authenticated_handshake (__main__.TestKeyExchange) ... ok
test_handshake_rejects_attacks (__main__.TestKeyExchange) ... ok
test_hkdf_rfc5869 (__main__.TestKeyExchange) ... ok
test_rfc7748_vectors (__main__.TestKeyExchange) ... ok
test_unauthenticated_exchange_is_open_to_mitm (__main__.TestKeyExchange) ... ok
test_x25519_matches_library (__main__.TestKeyExchange) ... ok

----------------------------------------------------------------------
Ran 19 tests in 13.011s

OK
tests run: 19, failures: 0, errors: 0, skipped: 0
```

*Screenshot: test run from my public key cryptography deliverable*

## What it shows

- RSA key pairs generated from scratch (2048, 3072 and 4096 bits)
- A simple PKI with a root CA, an issuing CA and signed certificates
- Secure key exchange: the handshake rejects attacks
- Key management: encrypted storage, expiry and rotation

# Catching up, then adding a new tool

## Active Directory DC

- Caught up on all domain controller work this semester
- Samba4 DC on an Ubuntu VM, rebuilt at home
- Completed and pushed to GitHub

## Entra ID

- Finished all pending Entra ID work
- Cloud identity side of AccessGuard Health in Azure
- Completed and pushed to GitHub
- Final stage: Conditional Access, and documenting everything

## Azure Key Vault

- Decided on as the project's new tool for secrets
- Implementation plan written and documented
- Uploaded to the learning folder

AZURE KEY VAULT

# Azure Key Vault: summary

A Microsoft Azure cloud service for securely storing and managing sensitive data: passwords, API keys, TLS/SSL certificates and cryptographic keys.

### Secrets management

Securely holds tokens, passwords and connection strings.

### Key management

Generates and controls the encryption keys that protect data.

### Certificate management

Provisions and manages SSL/TLS certificates, with automated renewals.

### Key security features

- Access control through Microsoft Entra ID, with Azure RBAC or fine-grained access policies
- Two tiers: Standard (software-protected) and Premium (FIPS 140-3 Level 3 hardware security modules)
- Soft-delete and purge protection guard against accidental or malicious deletion

### In AccessGuard Health

- Chosen as the project's new tool for secrets
- Plan written and uploaded to the learning folder
- Integration in progress

# Learning, FieldDAE and next steps

## Learning and networking

### Schoology videos

Watched the course videos to understand the topics more deeply and support my rubric work.

### FieldDAE event

Attended the FieldDAE event at DAE.

### GitHub portfolio

Every completed piece is documented and pushed to github.com/sabi-s/dae-6-month-Sabi.

## Next steps

- Finish Conditional Access and complete the project documentation
- Finish Configure Cryptographic Protocols (TLS, SSH keys, Wireshark analysis)
- Finish integrating Azure Key Vault into AccessGuard Health (due Nov 5)
- Add HashiCorp Vault as a future integration
- Keep applying for IAM and IT security analyst roles

LEARNING AND RESEARCH

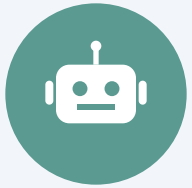
# AI-Assisted Research — Both Projects

I used three AI tools throughout Semester 1 and Semester 3 to research faster and think more clearly.



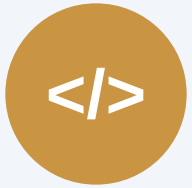
## Gemini NotebookLM

Organized research notes and source material, and generated grounded summaries of IAM & security concepts.



## ChatGPT

Brainstormed approaches, explained unfamiliar concepts, and helped debug Python logic.




## Claude

Supported technical planning, documentation, and structuring project write-ups and code.

***Takeaway:*** *AI tools sped up research and learning — every technical decision and every line of code was still mine.*

SEMESTER 5 | CLOSING

# Thank you

To my instructors, for your guidance, feedback and support throughout this program.

To my peers, for learning, building and sharing this journey with me.